

STRATA — one-page summary

Cleanverse Build: Trusted Assets Hackathon · DeFi track · Monad testnet · repo: github.com/Venkat5599/strata · live: strata-monad-nine.vercel.app

Problem

A compliance pool socializes ownership, so one non-compliant LP taints the whole pool — and the industry answer gates the pool in aggregate, so every LP pays the strictest restriction. Worse, Cleanverse's own `canTransfer` primitive answers non-compliance with a **hard revert**: an LP who legally owns 58% of a position can withdraw none of it.

Solution

STRATA moves the compliance boundary from the **pool** to the **position**: one balance, one price curve, N legal strata (OPEN / VERIFIED), and a withdrawal resolver that **grades instead of reverting** — returning `Direct`, `Routed`, or `Blocked` with the burnable/deferred split and the on-chain reason. Compliance partitions by credential; positions re-attribute when credentials change; the blocked branch prices at zero — so the **compliance basis** ($\text{basis}(a,b) = \text{price}(a) - \text{price}(b)$) becomes the first on-chain price of a transfer restriction.

- `previewExit(address, shares)` — pure view; grades any request, no wallet
- `deposit / depositAToken` — shares stamped with the depositor's credential
- `linkCredential` — migrates anonymous lots to the credential on upgrade
- `syncStratum` — permissionless mirror of the Policy pause state
- Four resolver invariants (conservation, no over-release, revocation absolute, freeze dominates) fuzzed at 10,000 runs; 48 tests green (33 local + 15 fork against the live Cleanverse contracts)

CVI · CVA integration points

Primitive	Role in STRATA	Live state
CVI · A-Pass 0xbA82D189...	<code>credentialOf()</code> reads the on-chain ERC-721 credential; positions key on the credential, never on <code>msg.sender</code>	Pool holds its own A-Pass (<code>balanceOf(pool) == 1, cvRecord 2089</code>); verified LP tier 50
CVA · A-Token	the registered instrument every policy question is denominated in; constructor rejects unregistered tokens	<code>isTokenRegistered(sCVA) == true</code>
Policy 0x36489bE4...	<code>policyClears()</code> wraps <code>canTransfer</code> (<code>revert → false</code>); <code>isPaused</code> drives the Blocked branch	aUSDC registered, policy unpaused — chips verified live on the dashboard
CVA · custodied	<code>depositAToken()</code> takes a registered A-Token directly	Live: pool custodies 250,000 sCVA — our own CVA, minted by us, real <code>depositAToken</code> (tx 0x7b489ad6)
Validator	registered via <code>POST /validator/register</code> with an EIP-191 owner signature	<code>is_register: true</code> , rules <code>min_tier: 1</code> , tx 0x983586fd
CCP export	<code>/api/ccp/export</code> — downloadable audit record of pool state + live credential	server route; API key never reaches the browser

Deployed

Chain: Monad testnet (10143) · RPC <https://testnet-rpc.monad.xyz>

Contract	Address	State
StrataPool	0x04df73761E1e524C0112D9a3633A44F8924BC31D	registered validator; OPEN 150,000 / VERIFIED 255,000
sCVA (our CVA)	0xa4C1B2d93D1F6A1cF83047C0C068ac15DEf7224f	Policy-registered; 250,000 custodied
A-Pass (CVI)	0xbA82D189540CaC9DC6FF46B6837CaC1BFdEC58B9	pool + verified LP hold credentials
Cleanverse Policy	0x36489bE45fa84f70a0c2BDB11D824Be608CB12Dd	aUSDC registered, unpaused
dUSDC (asset)	0x16CAf4d60BED18C215d1708870Ecc3fD9b46c242	open-mint demo dollar

App: strata-monad-nine.vercel.app (landing + section-based dashboard; every number a live contract read).

Honest notes: testnet USDC has no open mint and the USDC → aUSDC conversion is blocked upstream, so the OPEN stratum pools dUSDC and the custodied CVA is our own sCVA — both per Cleanverse's sanctioned guidance ("issuing your own CVA is permitted"). The pool architecture is unchanged for USDC-backed custody the moment their conversion pipeline works.